

**Set-up and registration**

# International Marketing Analytics



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[Office hours: book online \(Google Calendar\)](#)

# Today: from zero to a working set-up

Accounts

Tools

Repo

Project

Check

AI

## What you need



- Your laptop and charger
- Windows 10/11 or a recent macOS
- About 10 GB free disk space
- Admin rights to install software

## Have at hand



- Your WU e-mail and its password
- Your phone, for two-factor sign-in
- Your student ID or enrolment confirmation

## How we work



- One step at a time, all together
- Check your neighbour's screen
- Stuck? Raise your hand, keep going with the next step



**Goal:** by the end of today the check script prints “ready” on every laptop.

# Register your accounts



Part 1 of 6 · GitHub, GitHub Education,  
DataCamp

# Three registrations

## Accounts

## Tools

## Repo

## Project

## Check

## AI



### GitHub

Stores code and its history;  
hosts the course repository

[github.com/signup](https://github.com/signup)



### GitHub Education

Free student benefits,  
including GitHub Copilot

[education.github.com/pack](https://education.github.com/pack)



### DataCamp

Online Python courses; four  
certificates count 20 %

**Invitation link on Canvas**



**Same e-mail everywhere:** your WU student address unlocks the free plans.

## Accounts

## Tools

## Repo

## Project

## Check

## AI

1 Open [github.com/signup](https://github.com/signup)

2 **WU e-mail:** sign up with your student address

3 **Username:** short and professional, e.g. anna-berger

4 **Verify** the code GitHub sends to your inbox

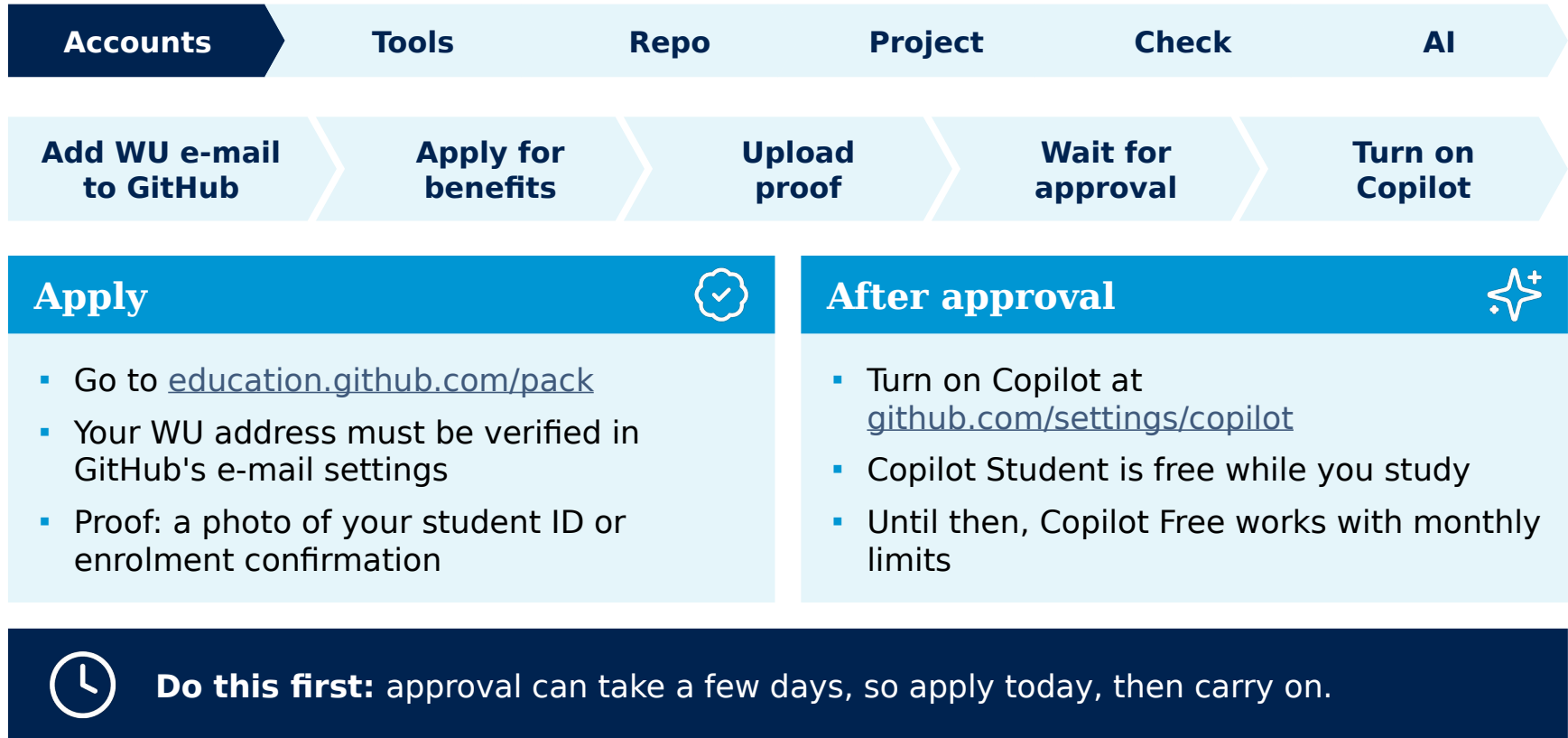
5 **Two-factor sign-in:** GitHub Mobile or an authenticator app

## Why it matters



- Your username appears on every commit you make
- A clean GitHub profile is part of your CV
- Optional: share your group's case study in a repository
- The free plan is all you need

# GitHub Education and Copilot



## Accounts

## Tools

## Repo

## Project

## Check

## AI

**1** **Open** the invitation link on Canvas

**2** **Sign up** with your WU e-mail, or log in

**3** **Join** the classroom International Marketing Analytics

**4** **Find** the four assigned courses under Assignments

**5** **Upload** each certificate to Canvas by its deadline

**6**

months free premium access

**4**

courses with certificates

**20 %**

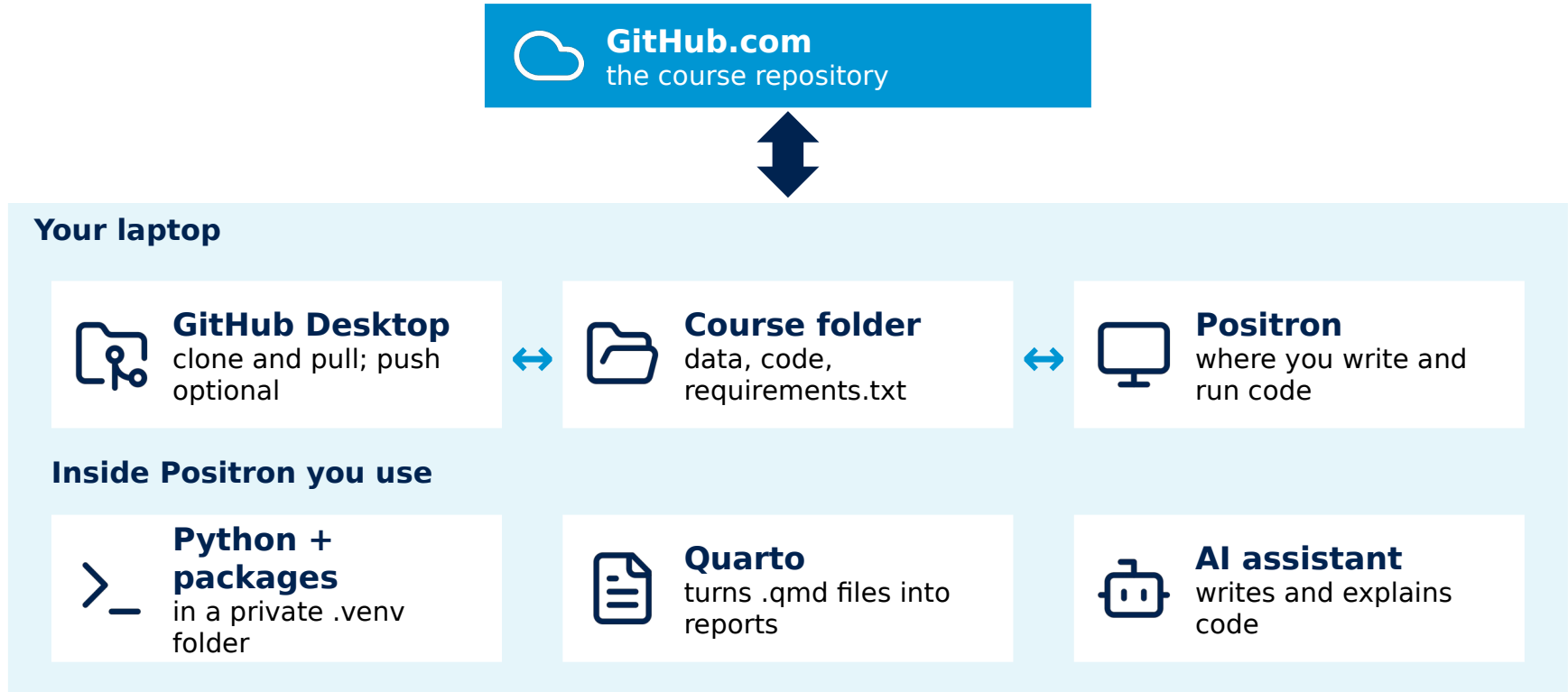
of the final grade

# Install the tools



Part 2 of 6 · Python, Positron, Quarto, GitHub  
Desktop, the course repository

# How the pieces fit together



## Step 2: install Python 3.14.8

Accounts

**Tools**

Repo

Project

Check

AI

### Windows



- **Download:** Python install manager
- Install it with a double-click
- Start menu → Terminal: `py install 3.14`

```
py -3.14 --version
```

### macOS



- **Download:** macOS installer (Python 3.14.8)
- Open the .pkg and click through
- Apple silicon and Intel

```
python3.14 --version
```



**Both downloads:** [python.org/downloads](https://python.org/downloads)

Or let Positron install it: Command Palette → Python: Install Python via uv

## Step 2: install Positron, Quarto and GitHub Desktop

Accounts

**Tools**

Repo

Project

Check

AI

### Positron



- The editor for code, data and charts
- Version 2026.09, free
- Welcome page checks that Python is ready

[positron.posit.co/download](https://positron.posit.co/download)

### Quarto



- Turns text and code into reports and slides
- Check in a terminal:  
quarto --version

[quarto.org/docs/get-started](https://quarto.org/docs/get-started)

### GitHub Desktop



- Clone and pull with clicks; push if your group shares a repository
- Installs Git for you
- Sign in with your GitHub account

[desktop.github.com](https://desktop.github.com)



**Restart Positron** after installing Quarto, so it finds it.

## Step 3: get the course repository

Accounts

Tools

**Repo**

Project

Check

AI

**Copy the URL  
(below)**

**GitHub Desktop:  
Clone repository**

**Pick a folder  
without spaces**

**Positron:  
Open Folder**

### Course repository (public)

[https://github.com/arnefloh-wu/  
international-marketing-analytics-teaching](https://github.com/arnefloh-wu/international-marketing-analytics-teaching)

**Folder:** e.g. C:\Users\anna\wu\ima or ~/wu/ima.  
Avoid spaces, umlauts and OneDrive or iCloud  
folders.

**Clone:** your own copy of a repository; a public  
one needs no sign-in

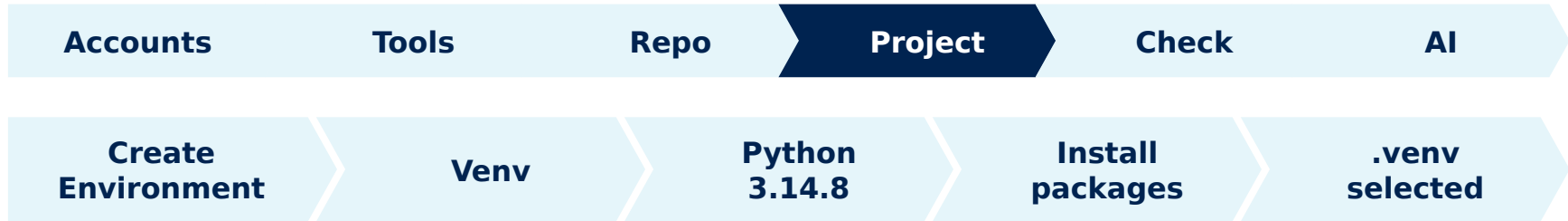
**Pull:** get the latest materials; do it before every  
session

**Commit and push:** optional: share work in a  
group repository



**In Positron:** File → Open Folder. The cloned folder is now your project.

## Step 4: set up the project's Python



### Positron then



- Start: Command Palette → Python: Create Environment
- Creates .venv inside the project
- Installs all course packages
- Starts Python from .venv

**Interpreter picker**  
top right

Python 3.14.8 (.venv) ▼

### Packages missing? Positron console

```
%pip install -r requirements.txt
```



**Once per project:** Positron remembers the .venv and starts it whenever you open the folder.

## Step 5: run the check

Accounts

Tools

Repo

Project

Check

AI

### Positron console (Windows and macOS)

```
%run setup/check_setup.py
```

### Example output (shortened)

```
Packages
[ok] polars          1.44.2    data wrangling
[ok] plotnine        0.15.8    charts
[ok] great_tables    1.0.0     tables
[ok] statsmodels      0.15.0     regression
...
ready
```

### Reading it



- [ok]: in place
- [!!!]: the line says what to fix
- Ends with “ready”: done



**Next:** the full test with test\_stack.qmd (next slide).

## Step 5: test the whole stack

### test\_stack.qmd · cells 1 and 2

```
# 1 Install: run once with Ctrl/Cmd + Enter
%pip install polars plotnine great-tables \
    statsmodels pyarrow jupyter
# 2 Load
import polars as pl
import statsmodels.formula.api as smf
from great_tables import GT
from plotnine import ggplot, aes, geom_line
```

### cells 3 and 4

```
# 3-4 polars and great_tables
sales = pl.read_csv(URL, try_parse_dates=True)
summary = (sales
    .filter(pl.col("week").dt.year() == 2025)
    .group_by("country")
    .agg(pl.col("revenue_eur_k").sum()))
GT(summary)
```

### cell 5

```
# 5 plotnine: weekly sales, AT and DE
two = sales.filter(
    pl.col("country").is_in(["AT", "DE"]))
ggplot(two, aes("week", "sales_units_k",
    color="country")) + geom_line()
```

### cell 6

```
# 6 statsmodels: regression for Austria
at = sales.filter(
    pl.col("country") == "AT").to_pandas()
model = smf.ols("sales_units_k ~ price_eur"
    " + spend_tv_k + spend_paid_search_k",
    data=at).fit()
```

**Result:** a table, a chart, the coefficients



**Run it:** cell 1 once with Ctrl/Cmd + Enter, then Preview (Ctrl/Cmd + Shift + K).

## Step 6: connect your AI assistant

Accounts

Tools

Repo

Project

Check

AI

- 1 Command Palette:** Configure Language Model Providers
- 2 Choose GitHub Copilot** and sign in with GitHub
- 3 Open the chat:** View: Show Posit Assistant

### Providers

**GitHub Copilot:** chat and completions; free with Copilot Student or Copilot Free

**Posit AI Pass:** chat and completions; about USD 20 a month

**Anthropic API key:** chat; pay as you go



**Our rule:** the AI writes, you check. Run the code, read the output and keep a short prompt log for each submission.

# How Positron works



Part 3 of 6 · panes, running code, projects, shortcuts, Posit Assistant

New ▼ Open ▼

case-study ▼

Python 3.14.8 (.venv) ▼

**1 Explorer**  
files of the open folder

data/  
analysis.py  
report.qmd  
requirements.txt

**2 Editor**  
write .py and .qmd files

# %%  
sales = pl.read\_csv(...)  
sales.head()

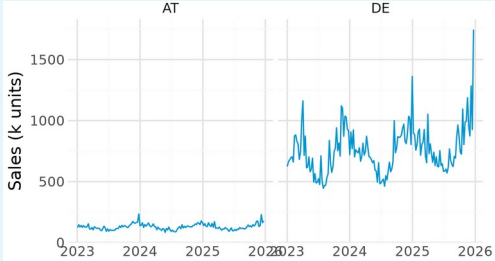
**3 Console and Terminal**  
the Panel: code runs, results print

>>> sales.height  
936

**4 Variables**  
Session pane: what is in memory

sales 936 × 18  
summary 6 × 3  
model OLS results

**5 Plots**  
Session pane: your charts



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**EQUIS**  
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ACCREDITED

ASSOCIATION  
**AMBA**  
ACCREDITED

# From code to result

**Write code  
in the editor**

**Ctrl/Cmd  
+ Enter**

**It runs in  
the console**

**Objects in  
Variables**

**Charts in  
Plots**

## Run code



- Ctrl/Cmd + Enter: the selection or current statement
- Ctrl/Cmd + Shift + Enter: the whole file, or the current cell
- # %% starts a cell in a .py file

## Look at results



- Variables: every object, its type and size
- Data Explorer: click a table, or type %view sales
- Plots: earlier charts, zoom, save as PNG



**Console or terminal?** The console speaks Python. The terminal speaks to your computer: pip, git and quarto commands go there.

# Projects in Positron: one folder for everything

## A project is a folder

```
case-study/
├── .venv/           private Python + packages
├── .git/            version history
├── .gitignore       what Git leaves out
├── data/            raw data, never edit by hand
├── analysis.py      code, split into # %% cells
├── report.qmd       text, code and results
└── requirements.txt packages and versions
```

**Open the folder**, not single files: File  
→ Open Folder

**Positron remembers** the interpreter  
and open files per folder

**Relative paths:** data/sales.csv works  
on every laptop

**Switch projects** with the project  
switcher, top right



**.venv stays on your laptop:** .gitignore keeps it out of GitHub. requirements.txt travels instead, so your group rebuilds the same environment with one command.

# Start a new project from a template

New Folder  
from Template

Python  
Project

Name and  
location

Python 3.14  
+ venv

Create

## Positron sets up



- The folder with a .venv inside
- Git: .git and a .gitignore
- A running Python session
- An empty file to start with

## Then, in the Positron console

```
%pip install polars plotnine great-tables
```

```
%pip freeze > requirements.txt
```

Line 1 installs packages, line 2 writes their versions to requirements.txt.



**Where:** New ▼ (top left), the project switcher, or Command Palette → Workspaces: New Folder from Template.

# Shortcuts worth knowing

| Action                             | Windows              | macOS               |
|------------------------------------|----------------------|---------------------|
| Run selection or current statement | Ctrl + Enter         | Cmd + Enter         |
| Run the file or current cell       | Ctrl + Shift + Enter | Cmd + Shift + Enter |
| Insert a code cell (# %%)          | Ctrl + Shift + I     | Cmd + Shift + I     |
| Restart the Python session         | Ctrl + Shift + 0     | Cmd + Shift + 0     |
| Clear the console                  | Ctrl + L             | Ctrl + L            |
| Preview a Quarto document          | Ctrl + Shift + K     | Cmd + Shift + K     |
| Command Palette                    | Ctrl + Shift + P     | Cmd + Shift + P     |
| Help for the word under the cursor | F1                   | F1                  |
| Accept an AI suggestion            | Tab                  | Tab                 |

# Posit Assistant: the AI inside Positron

| What it sees                                                      | What it does                                                                | How you steer it                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>▪ Your open files and project</li><li>▪ Loaded data, variables and plots</li><li>▪ Console history</li></ul> | <ul style="list-style-type: none"><li>▪ Explains, writes and fixes code</li><li>▪ Runs code after you approve</li><li>▪ Fix and Explain on any error</li></ul> | <ul style="list-style-type: none"><li>▪ <code>/plan</code>: a plan before any code</li><li>▪ <code>@file</code>: point it to a file</li><li>▪ Permission mode: keep Normal</li></ul> |

## Open it

View: Show Posit Assistant

Command Palette, or the Assistant icon in the Activity Bar

**Memory:** an AGENTS.md file in your project tells it your conventions, for example “use polars and plotnine”.

## Completions: ghost text



- Grey suggestions appear as you type
- Tab accepts, Ctrl/Cmd + → word by word
- Esc or keep typing to dismiss

## Chat: Posit Assistant



- Ask in plain English, one step at a time
- Check every result it reports
- Export the chat: your prompt log

## Good prompts

"Sum revenue by country for 2025 with polars and sort it."  
"Explain this error in two sentences, then fix it."



**Privacy:** prompts, variable names and types go to the model provider; rows of data only if you ask. Never paste personal data.

# Work with GitHub Desktop



Part 4 of 6 · the window, the everyday workflow, working as a group

# GitHub Desktop at a glance

**1** Current repository  
**ima-teaching** ▾

**2** Current branch  
**main** ▾

**3** Fetch origin  
↻ **now**

**4** **Changes (2)** History  

analysis.py

report.qmd

Add Austria sales chart

**Commit to main**

**5** summary, then commit

**6** **report.qmd**  

```
at = sales.filter(...)  
- at.head()  
+ ggplot(at, aes("week",  
+   "sales_units_k"))  
+   + geom_line()
```

**1** **Current repository**  
which project you work on

**2** **Current branch**  
main is all we need

**3** **Fetch / Pull / Push origin**  
sync with GitHub

**4** **Changes**  
files edited since last commit

**5** **Summary + Commit**  
save a snapshot with a message

**6** **Diff**  
red removed, green added

**Course repository:** you only pull. Work in your own folder; a group repository is optional.

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**AMBA**  
ACCREDITED

# The everyday workflow

**Fetch, then  
Pull origin**

**Copy files into  
your own folder**

**Work in  
Positron**

**Optional:  
commit, push**

## Course repository



- Required: pull before every session
- Read-only for you: copy files into your own folder before editing

## Group repository



- Optional: one member publishes a private repository
- Pull, work, commit, push
- One person per file where you can

## Good commits



- If you commit: small steps, often
- Summary says what changed: "Add AT sales chart"
- Push when you stop



**Merge conflict in a group repository?** Open the file, keep the lines you want, delete the markers (<<< === >>>), then commit and push.

# Working on the case study together

**Pull course  
repository**

**Copy into  
your folder**

**Positron:  
.venv**

**Combine:  
2 ways**

**Upload  
to Canvas**

## Project folder



- Copy data/case, report.qmd and requirements.txt
- report.qmd next to the data folder
- Create Environment from requirements.txt

## Way 1: simple



- One member keeps report.qmd
- The others send code chunks and text
- Render the combined file every week

## Way 2: GitHub



- Optional: Publish repository, private
- Invite on github.com: Collaborators
- Pull, commit, push



**Only Canvas counts:** upload report.qmd and report.html; your commit history is not assessed.

# Write reports with Quarto



Part 5 of 6 · how it works, the YAML header,  
code cells, output formats

# Quarto: text, code and results in one file



## report.qmd

```
---
title: "Alpenglow: sales in Austria"
author: "Anna Berger"
format: html
---

## Weekly sales

```{python}
at = sales.filter(pl.col("country") == "AT")
at.head()
```

Austria is our smallest market.
```

**1 YAML header:** settings between the two --- lines

**2 Markdown:** headings (##), **bold**, lists, links

**3 Code cells:** ```{python} ... ``` run Python; the output lands under the cell



**Preview:** Ctrl/Cmd + Shift + K in Positron. Render: quarto render report.qmd

# The YAML header: settings for the whole document

## Top of report.qmd

```
---
title: "Alpenglow: sales in Austria"
author: "Anna Berger"
date: today
format:
  html:
    toc: true
    code-fold: true
execute:
  warning: false
---
```

**title, author**

shown at the top

**date: today**

the day you render

**format: html**

the output; its options  
indented below

**toc: true**

table of contents

**code-fold: true**

code hidden behind a  
"Code" button

**warning: false**

no warnings in the output



**Indentation matters:** two spaces per level, no tabs, and a space after every colon.

# Code cells: options and numbers in the text

## Inside report.qmd

```
```{python}
#| label: fig-sales
#| fig-cap: "Weekly sales in Austria"
#| echo: false
ggplot(at, aes("week", "sales_units_k")) + geom_line()
```
```

The chart covers `{python} at.height` weeks (@fig-sales).

**#| label: fig-sales**  
names the cell; fig- makes it a figure

**#| fig-cap:**  
the caption under the chart

**#| echo: false**  
show the result, hide the code

**`{python} ...`**  
a computed number in the text

**@fig-sales**  
"Figure 1", with a link

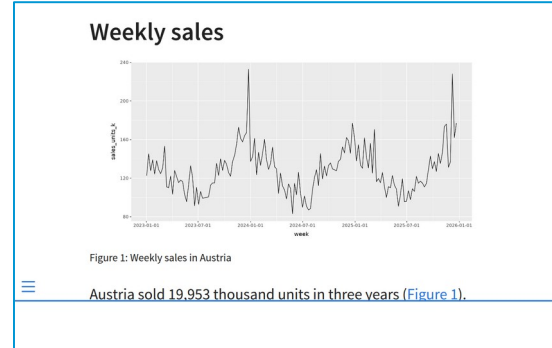


**Why it matters:** when the data change, you render again and every chart, table and number in the text updates. No copy and paste into Word.

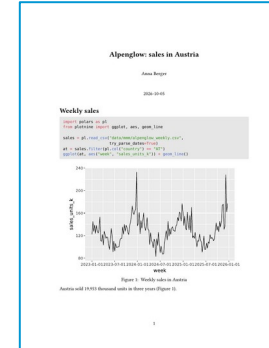
# One source, many output formats



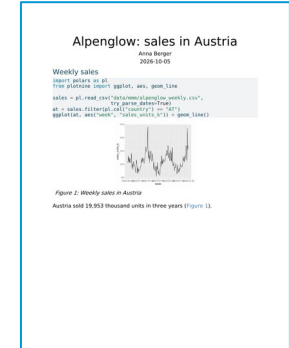
**html**  
web page (default)



**revealjs**  
slides in the browser



**typst**  
PDF, no LaTeX  
needed



**docx**  
Word, to edit further

## YAML: several formats

```
format:
  html: default
  revealjs: default
  typst: default
  docx: default
```

## Terminal: all formats, or one

```
quarto render report.qmd
quarto render report.qmd --to docx
```

Also: pptx, dashboard, pdf (LaTeX)

# Python packages for this course



Part 6 of 6 · install and load; polars, plotnine,  
Great Tables, statsmodels

# Install packages: three ways

## 1 Windows and macOS: Positron console (easiest)

```
%pip install polars plotnine great-tables
```

## 2 Windows: Positron terminal

```
.venv\Scripts\python -m pip install -r requirements.txt # all course packages  
.venv\Scripts\python -m pip install polars # one package
```

## 3 macOS: Positron terminal

```
.venv/bin/python -m pip install -r requirements.txt # all course packages  
.venv/bin/python -m pip install polars # one package
```



**Install name and import name can differ:** pip install great-tables → import great\_tables; scikit-learn → sklearn; pymc-marketing → pymc\_marketing.

# The Packages pane: install without code

**Packages icon  
(Activity Bar)**

**Install  
Package**

**Search name,  
pick version**

**Restart session  
when asked**

**PACKAGES** Python 3.14.8 (.venv)

|              |        |            |
|--------------|--------|------------|
| great-tables | 1.0.0  |            |
| pandas       | 3.0.6  | ● attached |
| plotnine     | 0.15.8 |            |
| polars       | 1.44.2 | ● attached |
| statsmodels  | 0.15.0 | ● attached |
| ...          |        |            |

## What else it does



- Shows which packages are loaded (attached)
- Update one, or Update All Packages
- Flags known security issues
- Keeps versions in line with requirements.txt
- Offers to install a missing package after an error

# Load packages: import at the top of every file

**Install once  
per laptop**

**Import in every  
file and session**

**Use the short  
name: `pl.read_csv()`**

## Windows and macOS: the same code

```
import polars as pl
import statsmodels.formula.api as smf
from plotnine import ggplot, aes, geom_line
from great_tables import GT

print(pl.__version__)    # 1.44.2
```

## Conventions

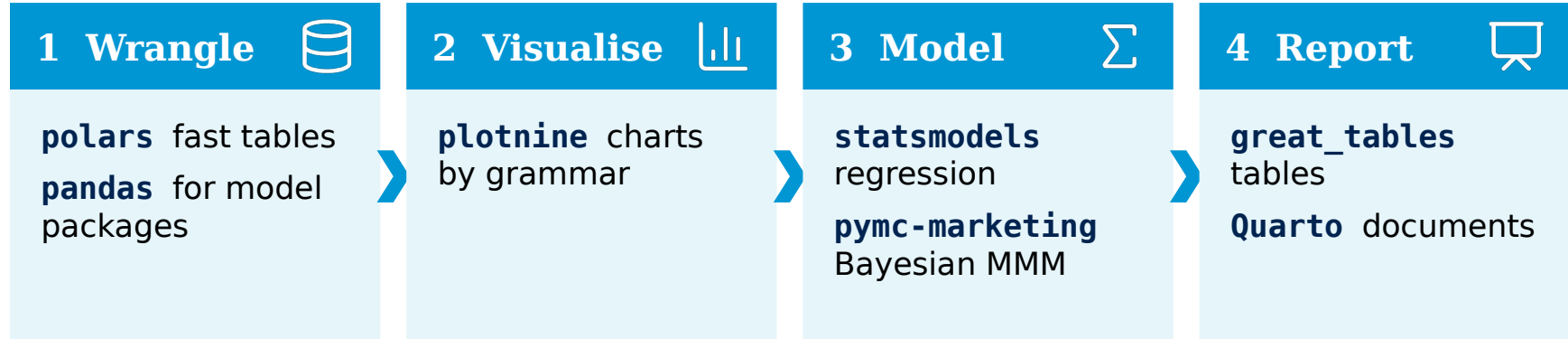


- `pl` and `smf` are the usual short names
- Imports go at the top
- Import only what you use



**ModuleNotFoundError?** Either the wrong interpreter is selected (pick the one in `.venv`) or the package is missing: `%pip install it`.

# Your toolkit, from data to report



**22**

packages pinned in requirements.txt

**1**

command installs them all

**0**

euros: all open source



**Online you will also meet pandas and matplotlib:** older but common; the ideas carry over.

## Revenue and TV spend per country, 2025

```
import polars as pl

sales = pl.read_csv("data/mmm/alpenglow_weekly.csv",
                    try_parse_dates=True)

summary = (
    sales
    .filter(pl.col("week").dt.year() == 2025)
    .group_by("country")
    .agg(pl.col("revenue_eur_k").sum().alias("revenue"),
         pl.col("spend_tv_k").sum().alias("tv"))
    .sort("revenue", descending=True)
)
```

**DataFrame:** a table with typed columns

**Expressions:** `pl.col("tv").sum()`  
says what to compute

**Chaining:** one step per line, read  
top to bottom

**Fast:** multi-threaded, written in Rust

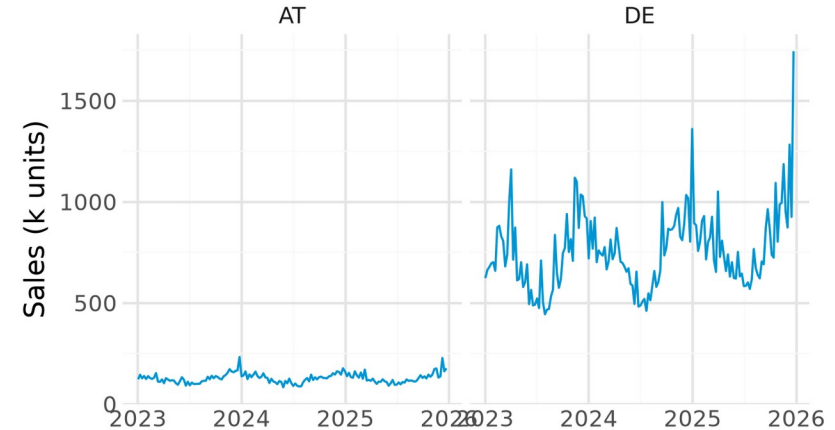


**Result:** six rows, one per country. Germany leads with about EUR 131 million revenue in 2025. Need pandas? Add `.to_pandas()`.

# plotnine: charts with the grammar of graphics

## Weekly sales, Austria and Germany

```
(ggplot(sales.filter(
  pl.col("country").is_in(["AT", "DE"])),
  aes(x="week", y="sales_units_k"))
+ geom_line(color="#0096D3")
+ facet_wrap("country")
+ scale_x_date(date_breaks="1 year",
  date_labels="%Y")
+ labs(x="", y="Sales (k units)")
+ theme_minimal())
```



The output of the code on the left

**ggplot()**  
the data

**aes()**  
columns to x, y

**geom\_**  
the marks

**facet\_**  
one panel per  
group

**theme\_**  
the look

**Data + mapping + marks + panels + theme:** each + adds one layer.

# Great Tables: tables for reports

## The polars summary as a table

```
from great_tables import GT

(GT(summary)
 .tab_header(title="Revenue and TV spend, 2025")
 .fmt_number(columns=["revenue", "tv"], decimals=0)
 .cols_label(country="Country",
              revenue="Revenue (k EUR)",
              tv="TV spend (k EUR)"))
```

**Header and labels:** readable names instead of column codes

**Number formats:** thousands separators, currency, percent

**Works with polars** and pandas; renders in Quarto reports

| Revenue and TV spend, 2025 |                 |                  |
|----------------------------|-----------------|------------------|
| Country                    | Revenue (k EUR) | TV spend (k EUR) |
| DE                         | 130,650         | 5,434            |
| FR                         | 66,306          | 2,363            |
| IT                         | 47,950          | 2,478            |
| NL                         | 33,234          | 812              |
| PL                         | 23,112          | 616              |
| AT                         | 21,984          | 695              |

## Weekly sales in Austria, 2023 to 2025

```
import statsmodels.formula.api as smf

at = sales.filter(pl.col("country") == "AT").to_pandas()
model = smf.ols(
    "sales_units_k ~ price_eur + spend_tv_k"
    " + spend_paid_search_k",
    data=at).fit()
print(model.summary())
```

| Coefficient              | Estimate |
|--------------------------|----------|
| Intercept                | 250.63   |
| price_eur                | -48.31   |
| spend_tv_k               | 0.41     |
| spend_paid_search_k      | 4.03     |
| R <sup>2</sup> (n = 156) | 0.43     |



**Reading it:** EUR 1,000 more paid search in a week goes with about 4,000 more units sold, holding price and TV constant.

**Later in the course:** scikit-learn for logistic regression and pymc-marketing for Bayesian MMM.

# When something goes wrong

| Problem                                        | Fix                                                                        |
|------------------------------------------------|----------------------------------------------------------------------------|
| <b>Windows: python or py is not recognised</b> | Reinstall the Python install manager, then run py install 3.14             |
| <b>No .venv in the Interpreter picker</b>      | Open the course folder, not a single file; then pick Python 3.14.8 (.venv) |
| <b>ModuleNotFoundError</b>                     | Select the .venv interpreter, then %pip install the package                |
| <b>macOS: externally-managed-environment</b>   | You used the system Python: select the project's .venv first               |
| <b>Quarto preview does not start</b>           | Restart Positron; run quarto check in the terminal                         |
| <b>Clone fails in GitHub Desktop</b>           | Sign in again under Accounts in the settings; check the URL                |
| <b>Copilot is not offered</b>                  | GitHub Education still pending: use Copilot Free meanwhile                 |

# Checklist: before you leave today

## Accounts



- ☐ GitHub account with two-factor sign-in
- ☐ GitHub Education application sent
- ☐ DataCamp classroom joined

## Laptop



- ☐ Python 3.14, Positron, Quarto, GitHub Desktop
- ☐ Course repository cloned and open
- ☐ .venv created and selected
- ☐ “ready” and the test report renders
- ☐ Posit Assistant connected



**Not ready yet?** Post the failing step in the Q&A forum on Canvas before Session 2.



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